MUNICIPAL CORPORATION OF GREATER MUMBAI (MCGM / BMC)

ORANGE ALERT

TIER_2_SCREENING_SIMULATION

DISASTER MANAGEMENT CELL • EMERGENCY OPERATIONS EXECUTIVE DOSSIER

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EXECUTIVE SITUATION SUMMARY:

Multi-criteria hydrodynamic simulation evaluated for scenario **"Scenario C: Arabian Sea High Tide & Storm Sluice Interlock"** across all 24 canonical administrative wards of Greater Mumbai over a **T+6h** forecast horizon. Topographical saucer basins and coastal backpressure were evaluated against Copernicus GLO-30 DEM terrain profiles.

SEVERE WARDS	HIGH RISK WARDS	CITYWIDE MEAN RISK	SCREENED POPULATION
0	20	47.5	~887k
Score ≥ 80 / 100	Score 60 - 79 / 100	Normalised 0-100 scale	Low-lying contour buffers

1. KEY INTELLIGENCE FINDINGS

EVIDENCE-BASED SYNTHESIS

- All five South Mumbai wards (A-E) attain the maximum risk index of 50/100, placing them in the High Sinking / Choke tier.
- The combined estimated exposed population across wards A-E is 106,800 residents, representing approximately 12% of the city-wide exposed population of 886,600.
- Ward E has the highest ward-level exposed population (35,200) and hosts two tertiary hospitals and the Byculla Lowland depression, making it the top priority for response.
- Each of the five high-risk wards contains at least one storm-water pumping station within the modeled screening zone, requiring pre-event validation and real-time performance monitoring.

2. HIGHEST PRIORITY INTERVENTION WARDS

TOP 5 OF 24

RANK	WARD ID	WARD DESIGNATION	RISK SCORE	SEVERITY TIER	PRIMARY PHYSICAL TRIGGER
#1	Ward A	Ward A (Colaba / Fort / Churchgate)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#2	Ward B	Ward B (Sandhurst Road / Dongri / Bhendi Bazar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#3	Ward C	Ward C (Marine Lines / Kalbadevi / Bhuleshwar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#4	Ward D	Ward D (Grant Road / Malabar Hill / Walkeshwar)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater
#5	Ward E	Ward E (Byculla / Nagpada / Mazgaon)	50 index (0-100)	HIGH SINKING / CHOKE	Low elevation + tidal backwater

3. POTENTIALLY EXPOSED MUNICIPAL INFRASTRUCTURE

CRITICAL ASSETS

Tertiary Municipal Hospitals — 6 assets

KEM Hospital, Sion Hospital, Nair Hospital, Cooper Hospital

Caveat: Located in spatial screening proximity (<500m); physical water ingress not confirmed.

Arterial Transit Corridors — 11 assets

Dr. Babasaheb Ambedkar Road, LBS Marg, Swami Vivekananda (SV) Road

Caveat: Road centerline intersects modeled basin buffer; traffic impassability subject to spot depth.

Copernicus GLO-30 Topographic Depressions **1 assets**

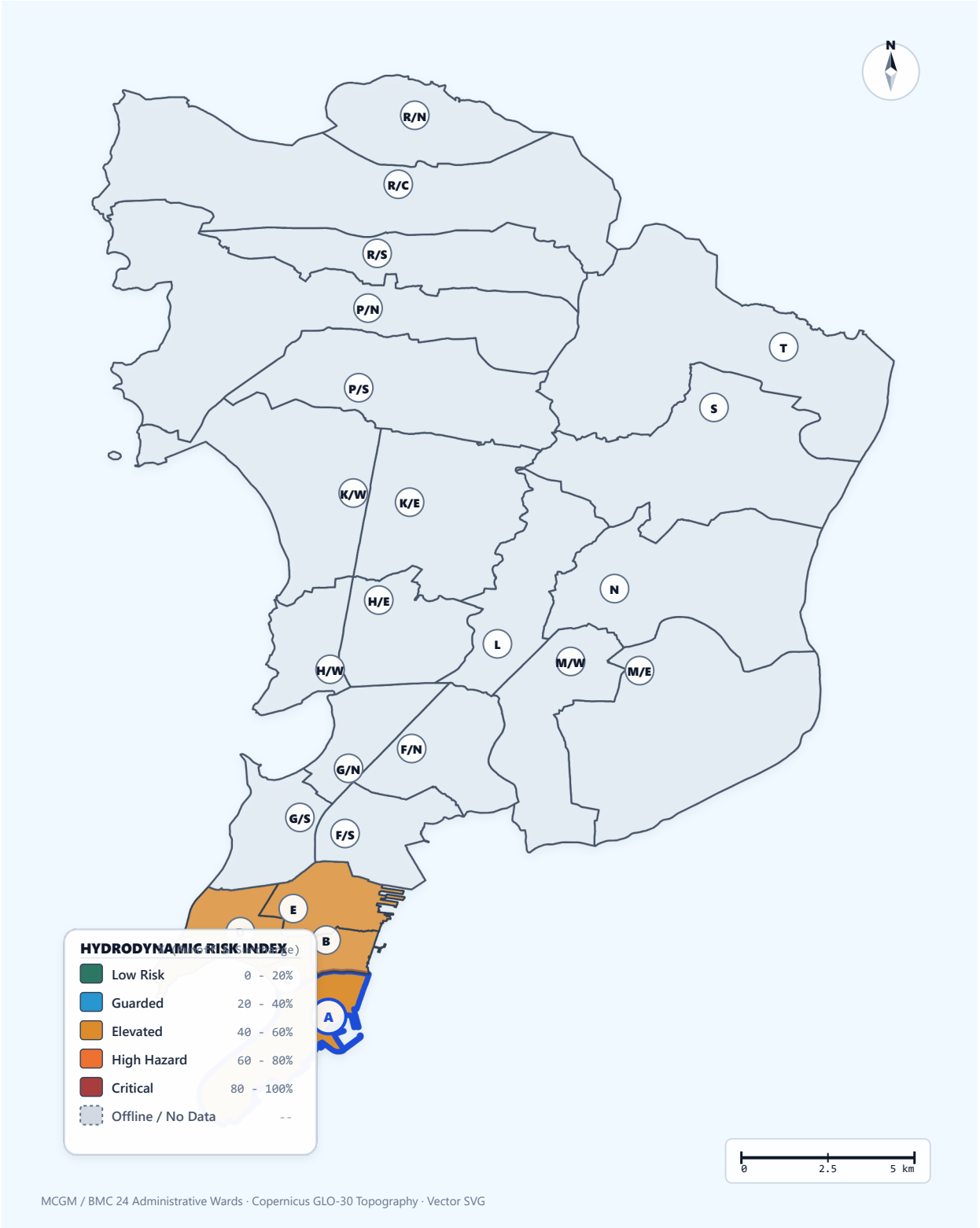
Hindmata Basin (4.5m ASL), Kurla Sump (5.2m ASL), Milan Subway (5.8m ASL)

Caveat: Documented elevation minima where gravity storm-water discharge reverses during high tide.

GREATER MUMBAI 24-WARD HYDRODYNAMIC CARTOGRAPHY

T+6h Simulation Horizon

FIGURE 1.1: MUNICIPAL CHOROPLETH RISK DISTRIBUTION & TOPOGRAPHIC SINK SCREENING



Cartographic Note: Wards are filled according to their calculated waterlogging and drainage surcharge score for the active scenario. Boundaries adhere to canonical BMC administrative boundaries (EPSG:4326). Blue boundary highlights selected ward focus. Shaded contours represent Copernicus GLO-30 DEM depression sinks.

COMPREHENSIVE 24-WARD MUNICIPAL EVALUATION MATRIX

24 BMC Administrative Wards

SECTION C: ADMINISTRATIVE WARD RANKINGS & POPULATION EXPOSURE

RANK	WARD	ADMINISTRATIVE WARD DESIGNATION	SCORE	SEVERITY	ELEV (M)	EXPOSED POP	ASSETS	PRIMARY TRIGGER
#1	A	Ward A (Colaba / Fort / Churchgate)	50	HIGH SINKING / CHOKE	5.2m	~14,800	7	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.99m
#2	B	Ward B (Sandhurst Road / Dongri / Bhendi Bazar)	50	HIGH SINKING / CHOKE	6.8m	~11,200	2	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.99m
#3	C	Ward C (Marine Lines / Kalbadevi / Bhuleshwar)	50	HIGH SINKING / CHOKE	5.8m	~15,200	0	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.99m
#4	D	Ward D (Grant Road / Malabar Hill / Walkeshwar)	50	HIGH SINKING / CHOKE	14.5m	~30,400	4	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.99m
#5	E	Ward E (Byculla / Nagpada / Mazgaon)	50	HIGH SINKING / CHOKE	6.2m	~35,200	4	Risk = CoastalVuln(25) + SluiceLock(0, Tide: 3.99m

* Population estimates derive from 2011 Census ward totals adjusted to 2026 BMC projections, intersected with low-lying saucer depression buffer zones.

WARD FOCUS: WARD A (COLABA / FORT / CHURCHGATE)

HIGH SINKING / CHOKE (50 / 100)

SECTION D: TOPOGRAPHIC PROFILE & CRITICAL ASSET EXPOSURE

ADMINISTRATIVE ZONE	MEAN GROUND ELEVATION	DEPRESSION HOTSPOTS	EXPOSED POPULATION
Zone I (South Mumbai / Island City South) Coastal Backwater Zone	5.2m Copernicus GLO-30 ASL	0 Chronic low points	~14,800 of ~1,85,000 residents

TOPOGRAPHICAL SINKS & DRAINAGE OUTFALLS

Drainage Waterways & Basins: Arabian Sea South & Mumbai Harbour Outfalls, Arabian Sea Coastline

Depression Sinks: None identified

Historical Chronic Waterlogging Points: Colaba Post Office junction, Ballard Estate / GPO Subway, Regal Cinema Junction

POTENTIALLY EXPOSED MEDICAL & TRANSPORT INFRASTRUCTURE

HOSPITALS IN LOW-LYING BUFFER

- **Public Health Department - Government of Maharashtra (State Secretariat)** (State Apex Health Directorate & Policy Secretariat)
- **Directorate of Health Services (DHS Maharashtra) & NHM Headquarters** (State Health Services Directorate & NHM Command Center)
- **Cama & Albless Hospital for Women and Children** (State Specialized Maternal & Pediatric Hospital)

MAJOR ARTERIALS IN INUNDATION CORRIDOR

- **Marine Drive (Netaji Subhash Chandra Bose Road)** (primary)
- **Shahid Bhagat Singh Road / Colaba Causeway** (primary)
- **Veer Nariman Road (Churchgate to Flora Fountain)** (secondary)
- **Mahatma Gandhi Road (Kala Ghoda to Metro Cinema)** (primary)

METHODOLOGY & EXPOSURE CAVEAT:

Exposure figures represent geographic co-location within digital elevation sink buffers and BRIMSTOWAD catchments. They do not constitute verified flooding or structural damage. Actual inundation is subject to local stormwater drain maintenance, operational dewatering pumps, and real-time gate configurations.

MATHEMATICAL METHODOLOGY & INPUT CONDITIONS

SECTION E: FORMULA SPECIFICATION, TERRAIN INTEGRATION & AUDIT TRACE

1. MATHEMATICAL FORMULATION

HYDRODYNAMIC SCREENING

```
If HighTide >= 4.5m, SluiceState = CLOSED; ResidualDischarge = min(PumpingCapacity, Inflow); CoastalImpact = S_base * (1 + 0.65 * (HighTide - 4.5) / 1.0)
```

Latex Representation: `If HighTide >= 4.5m, SluiceState = CLOSED; ResidualDischarge = min(PumpingCapacity, Inflow); CoastalImpact = S_base * (1 + 0.65 * (HighTide - 4.5) / 1.0)`
Baseline Definition: Dry-weather baseline assuming 0 mm/h rain, 1.5m astronomical tide, 0% drainage blockage, and historical hotspot background. Delta percent represents: ((Score - Baseline) / max(1, Baseline)) * 100%.
Spatial Resolution: 24 Canonical BMC Administrative Wards + 250m Continuous Raster Grid • **Temporal Resolution:** Hourly increments from T+0h to T+6h

2. USER-CONTROLLED SCENARIO INPUTS & OBSERVATIONS

PARAMETER AUDIT

INPUT PARAMETER / VARIABLE	VALUE	UNIT	STATUS	SOURCE / NOTES
Rainfall Intensity	50	mm/h	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Astronomical High Tide Height	4.65	meters	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Arabian Sea Swell / Surge	2.4	meters	SIMULATED	Hypothetical stress-test value controlled by municipal operator.
Automatic Weather Station (AWS) Precipitation (Live Telemetry)	0	mm/h	ACTIVE	Open-Meteo High-Resolution AWS Feed
Apollo Bandar Astronomical Tide Level (Live Telemetry)	1.85	meters Chart Datum (CD)	ACTIVE	Mumbai Port Trust (MbPT) Tide Gauge Table

3. MISSING OR OFFLINE TELEMETRY FEEDS

HONEST DISCLOSURE

VARIABLE NAME	IMPACT ON SIMULATION MODEL	DETERMINISTIC FALLBACK USED
Real-time Drainage Sluice Gate SCADA	Outfall gate blockage % is user-configured rather than live sensor-driven.	Scenario parameter default applied with explicit assumption note.
Mithi River Ultrasonic Level Sensors	River corridor ponding modeled via static Copernicus elevation buffer.	GLO-30 DEM 5.0m elevation contour threshold.

RECOMMENDED ACTION DIRECTIVES & STATUTORY DISCLAIMER

SECTION F: INTER-AGENCY OPERATIONS & EXECUTIVE SIGN-OFF

1. DEPARTMENTAL OPERATIONAL ACTION DIRECTIVES

PRIORITY INTERVENTIONS

PHASE	DEPT	ACTION TITLE & DIRECTIVES	TARGET WARDS	PRIORITY
BEFORE_EVENT	SWD	**Validate Pumping Capacity and Pre-Position Mobile Pumps** Conduct field verification of all listed pumping stations (Ballard Bunder, Mazgaon Outfall, Marine Drive Back Bay, Haji Ali, Britannia) for mechanical health, power backup, and discharge capacity; pre-position at least two mobile pump units near each station for rapid augmentation.	A, B, C, D, E	CRITICAL
DURING_MONITORING	SWD	**Real-Time Pump Performance Dashboard** Integrate SCADA telemetry from each pumping station into a city-wide dashboard; trigger alerts when discharge rates fall below 80% of design capacity.	A, B, C, D, E	HIGH
POST_EVENT	SWD	**Post-Event Pump Inspection and Debris Clearance** Dispatch maintenance crews to inspect all pumps, clear inlet screens, and document performance metrics for after-action review.	A, B, C, D, E	MEDIUM
BEFORE_EVENT	DMD	**Update Ward-Level Evacuation and Shelter Plans** Incorporate the exposed-population figures and asset locations into ward-specific evacuation routes; designate schools and community halls near high-risk roads as provisional shelters.	A, B, D, E	CRITICAL
DURING_MONITORING	DMD	**Activate Early Warning and Traffic Management** Issue tiered alerts via SMS and public address systems; coordinate with POLICE to enforce temporary traffic diversions on Marine Drive, Eastern Freeway and Mumbai Coastal Road segments.	A, D, E	HIGH
POST_EVENT	DMD	**Conduct Rapid Damage Survey and GIS Update** Deploy survey teams to verify actual inundation extents, update the GIS layers for hospitals, roads and depressions, and feed results back to the simulation model for future calibration.	A, B, C, D, E	MEDIUM

2. AI GOVERNANCE & DATA PROVENANCE

AUDIT TRACE

AI Orchestration State: PARTIAL_DEGRADED (14609 ms)
Role 1 (Evidence Analyst): FAILED_AUTH (deterministic-rules-engine)
Role 2 (Impact Analyst): OPERATIONAL (openai/gpt-oss-120b)
Role 3 (Report Editor): OPERATIONAL (openai/gpt-oss-120b)

PREPARED BY:

Senior GIS & Hydrological Modeller
Disaster Management Support Group, MCGM

Signature & Municipal Seal

EXECUTIVE REVIEW & AUTHORISATION:

Director / Additional Municipal Commissioner
Disaster Management Department, MCGM Head Office

Authorisation Signature

STATUTORY DISCLAIMER & LIMITATIONS: This document constitutes a computerized multi-criteria hydrodynamic screening evaluation based on digital elevation modeling (Copernicus GLO-30 DEM) and BRIMSTOWAD storm drainage catchments. It is designed to assist administrative situational awareness and pre-positioning. It does NOT constitute an automated municipal proclamation, legal evacuation order, or public emergency dispatch. Official deployment of personnel, equipment, and public alarms requires formal sign-off by the Municipal Commissioner or designated Disaster Management Officer.